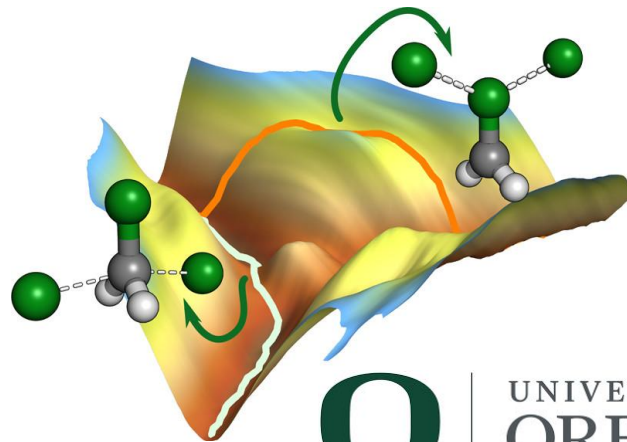
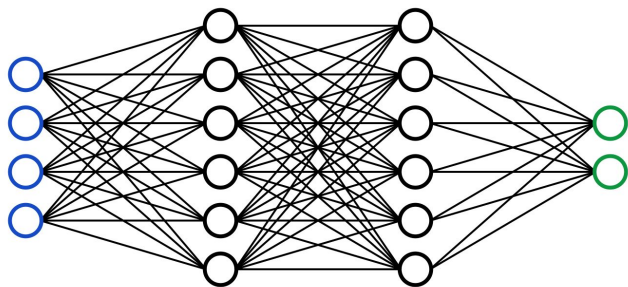


Machine Learning for Chemistry

Winter 2025

Topic 7: Clustering



UNIVERSITY OF
OREGON

Instructor: Dr. Dhiman Ray

Assistant Professor, Department of Chemistry and Biochemistry

Overview of Topics

- K-means Clustering
- K-medoids Clustering
- Assessing optimal number of clusters
- Agglomerative Clustering
- DBSCAN

References:

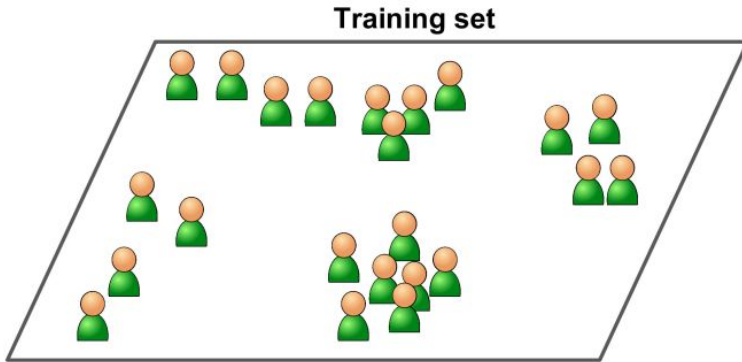
1. **Machine Learning with Pytorch and Scikit-Learn** by Raschka, Liu, and Mirjalili (Chapter 10)

Unsupervised Learning

Finding patterns in unlabelled training data.

1. Dimensionality Reduction

2. **Clustering**



Supervised learning

- > Labeled data
- > Direct feedback
- > Predict outcome/future

Unsupervised learning

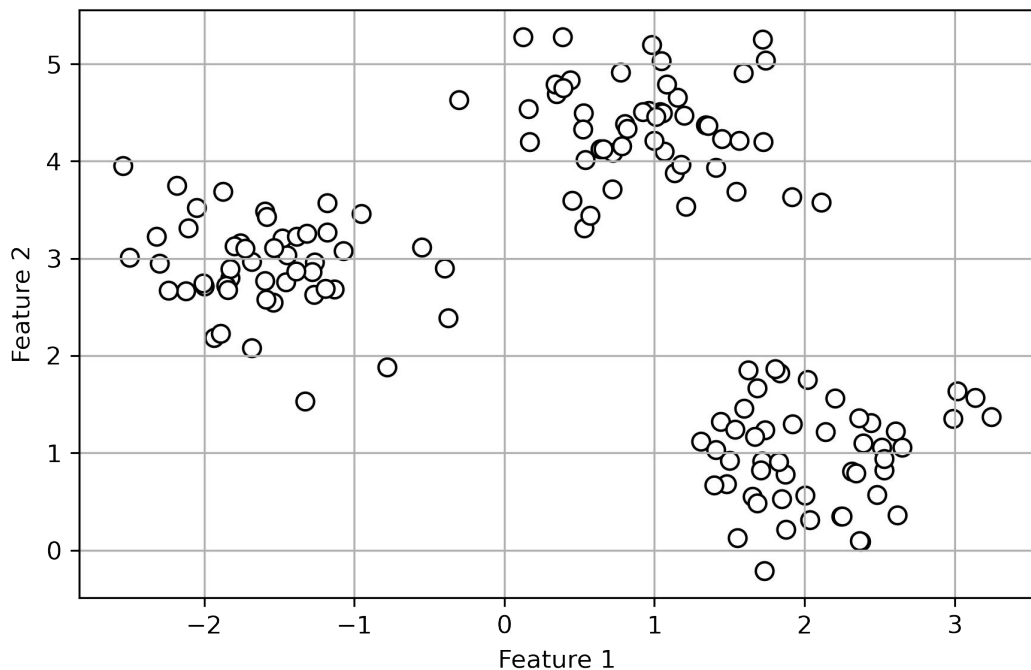
- > No labels/targets
- > No feedback
- > Find hidden structure in data

Reinforcement learning

- > Decision process
- > Reward system
- > Learn series of actions

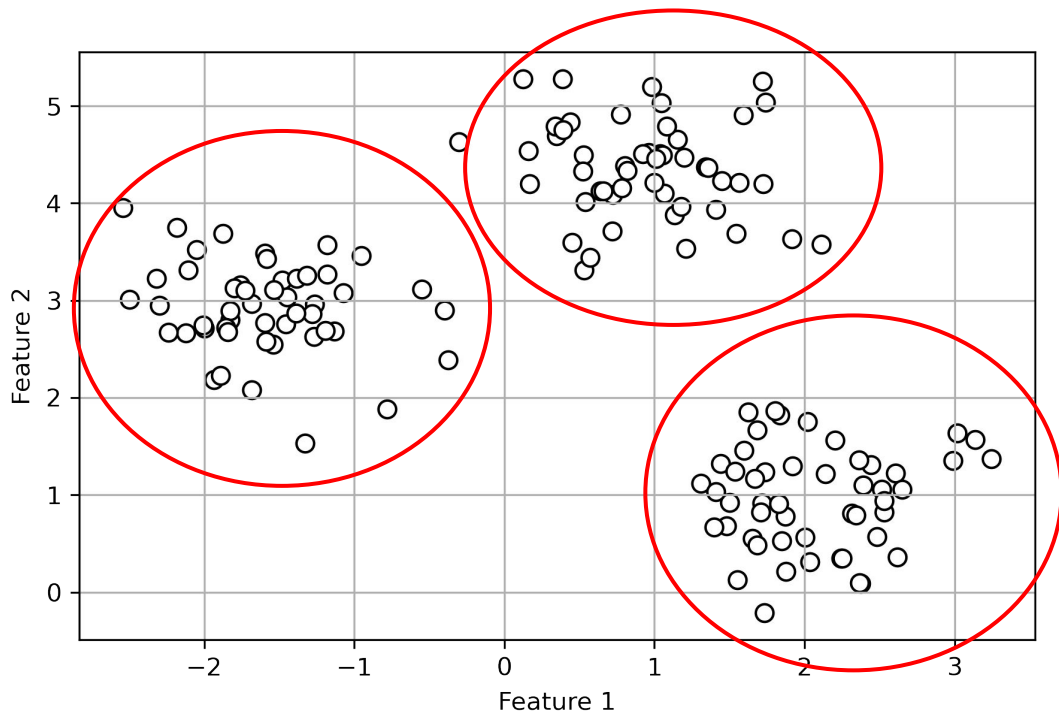
Unsupervised Learning: Clustering

Clustering aims to find a natural grouping in data so that items in the same cluster are more similar to each other than to those from different clusters.



Unsupervised Learning: Clustering

Clustering aims to find a natural grouping in data so that items in the same cluster are more similar to each other than to those from different clusters.



k-means Clustering



k-means Clustering

Step 1: Choose the number of clusters ($k=2$)

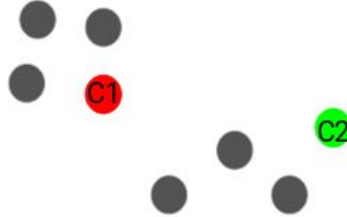


k-means Clustering

Step 1: Choose the number of clusters ($k=2$)



Step 2: Select k random points from the data as centroids

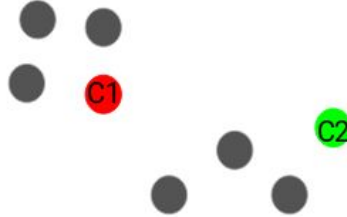


k-means Clustering

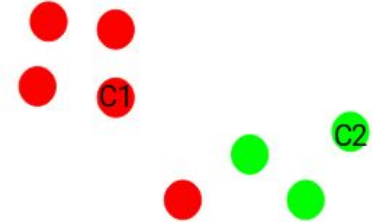
Step 1: Choose the number of clusters ($k=2$)



Step 2: Select k random points from the data as centroids



Step 3: Assign all the points to the closest cluster centroid

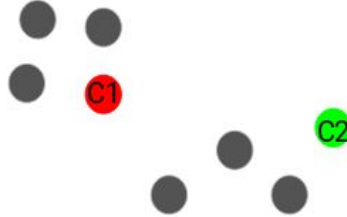


k-means Clustering

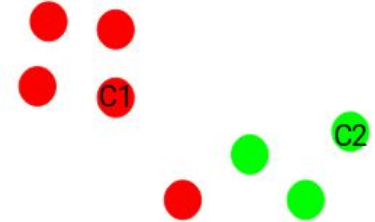
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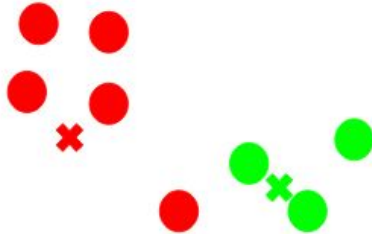
Step 2: Select k random points from the data as centroids



Step 3: Assign all the points to the closest cluster centroid



Step 4: Recompute the centroids of newly formed clusters

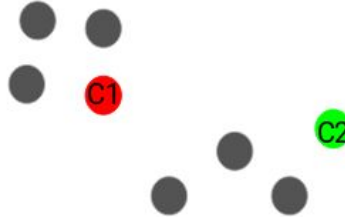


k-means Clustering

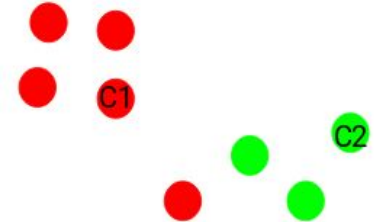
Step 1: Choose the number of clusters ($k=2$)



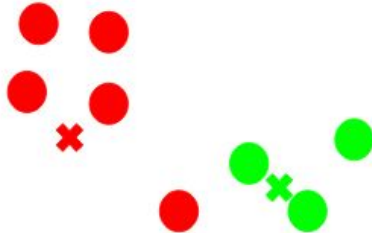
Step 2: Select k random points from the data as centroids



Step 3: Assign all the points to the closest cluster centroid



Step 4: Recompute the centroids of newly formed clusters



Step 5: Repeat steps 3 and 4 until (a) Centroids of newly formed clusters do not change or (b) Points remain in the same cluster or (c) Maximum number of iterations is reached

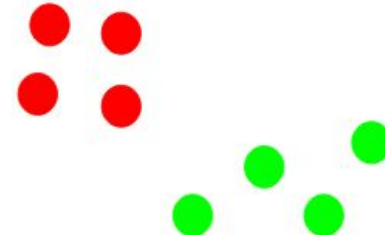


Figure courtesy: Pulkit Sharma 2024

k-means Clustering

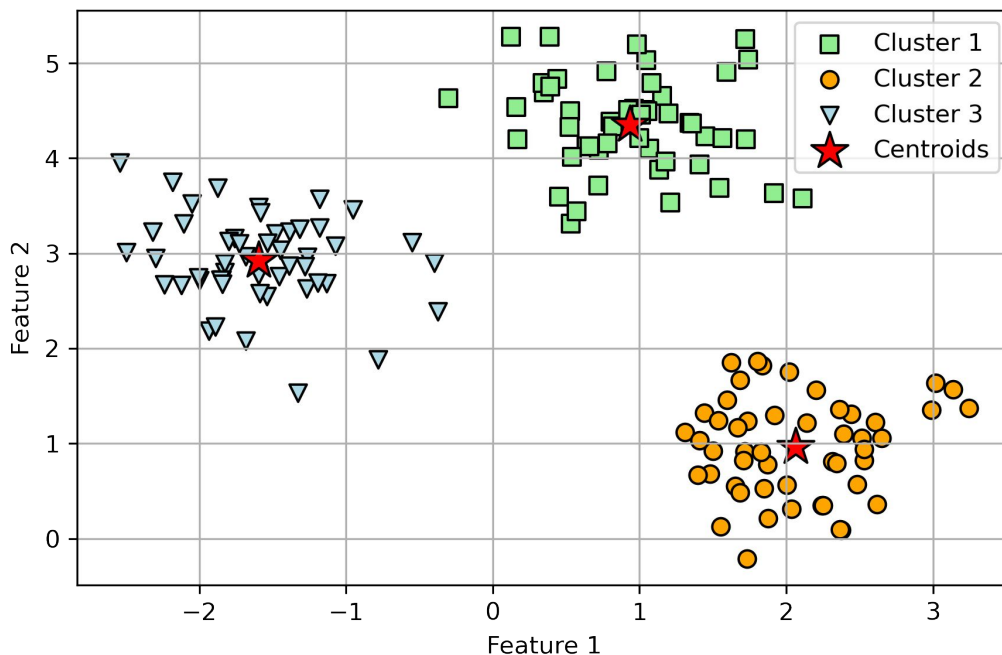
Distance measure:

Euclidean distance

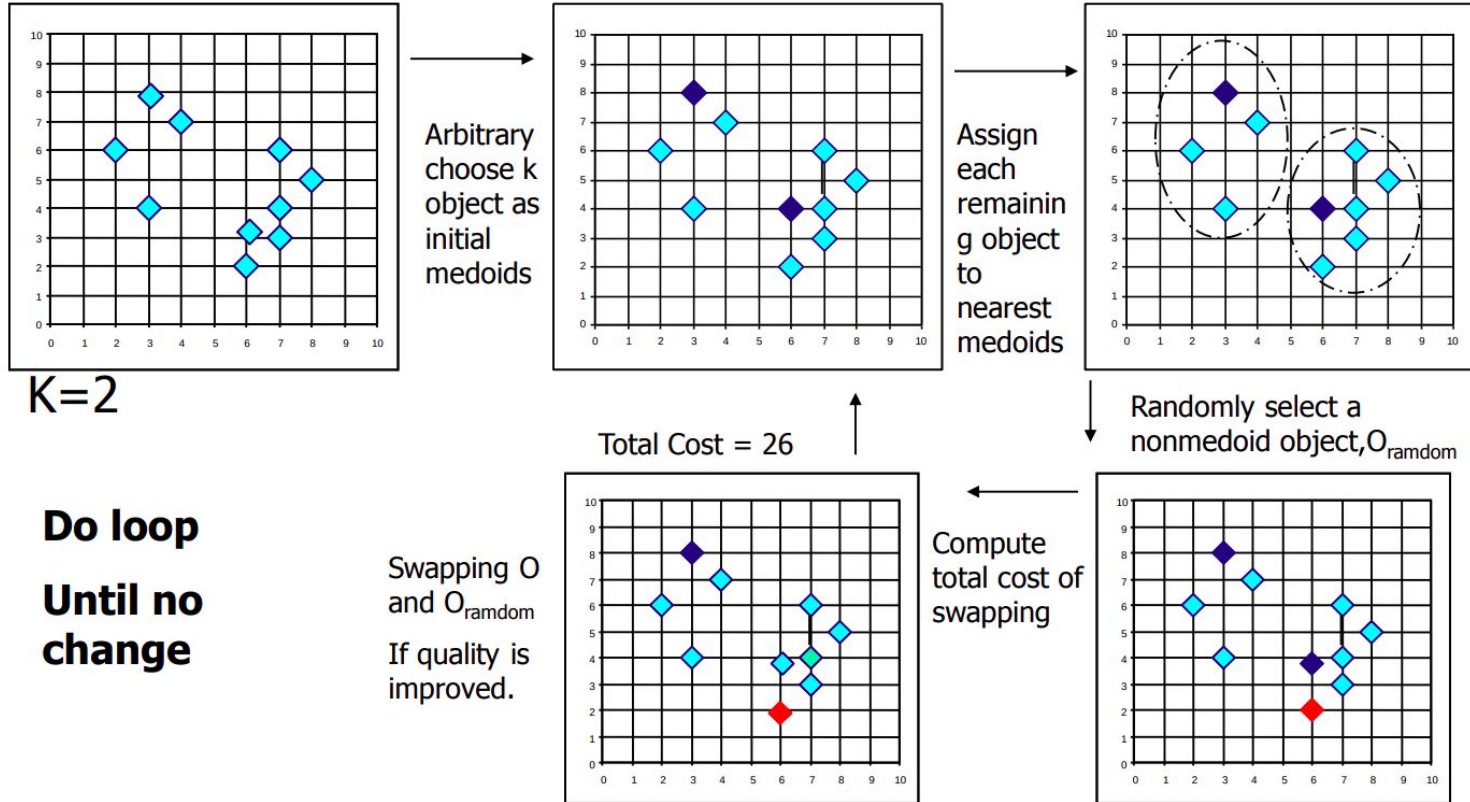
$$d(\mathbf{x}, \mathbf{y})^2 = \sum_{j=1}^m (x_j - y_j)^2 = \|\mathbf{x} - \mathbf{y}\|_2^2$$

Limitations:

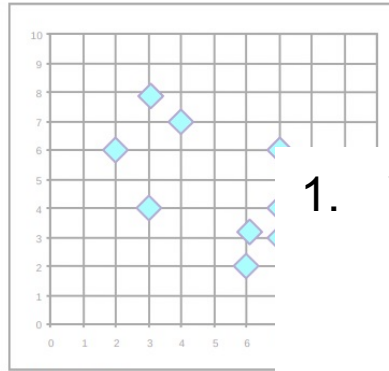
1. We have to choose appropriate number of clusters.
2. The cluster center is a fictitious point and not a member of the original dataset



k-medoids Clustering: PAM algorithm



k-medoids Clustering: PAM algorithm

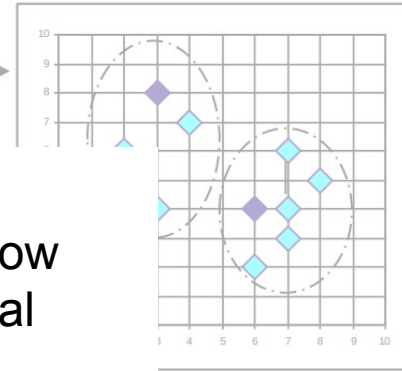


K=2

Arbitrary



Assign



1. We solve the problem of cluster centers being fictitious points. Now cluster centers (medoids) are real data.

2. We still have the problem of arbitrarily choosing the number of clusters.

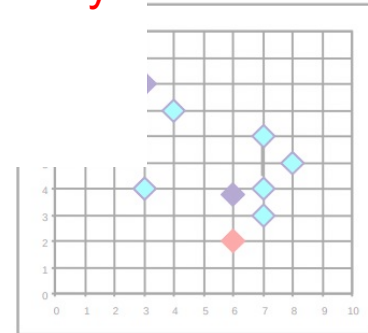
Do loop

Until no
change

If quality is
improved.



only select a
medoid object, O_{random}



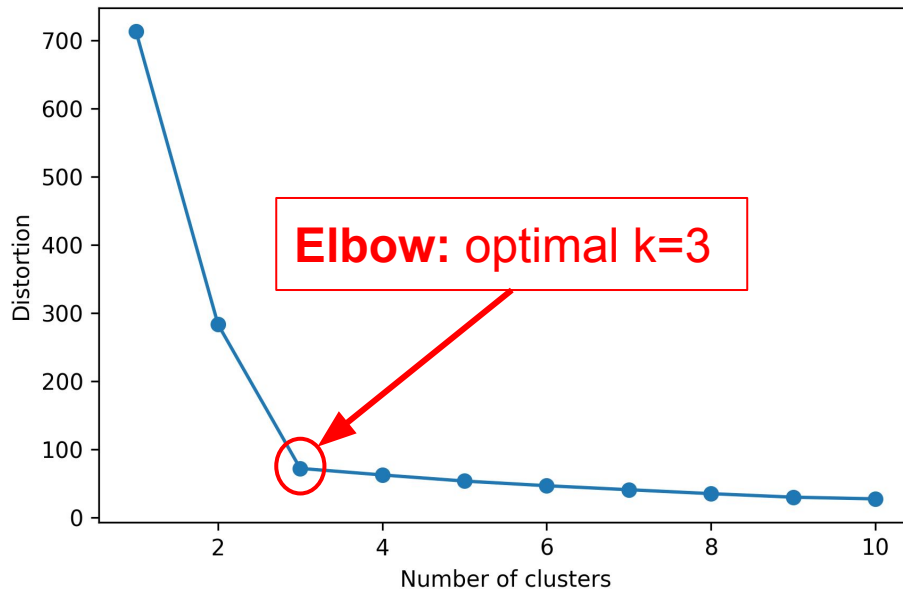
How to choose the optimal number of clusters?

Sum Squared Error (SSE)

$$SSE = \sum_{i=1}^n \sum_{j=1}^k w^{(i,j)} \|x^{(i)} - \mu^{(j)}\|_2^2$$

$\mu^{(j)}$ is the representative point (centroid) for cluster j . $w^{(i,j)} = 1$ if the example, $x^{(i)}$, is in cluster j , or 0 otherwise.

Elbow method: With increasing number of cluster (k) the SSE (distorsion) will decrease. After we reach the optimal k , the decrease will not be significant.



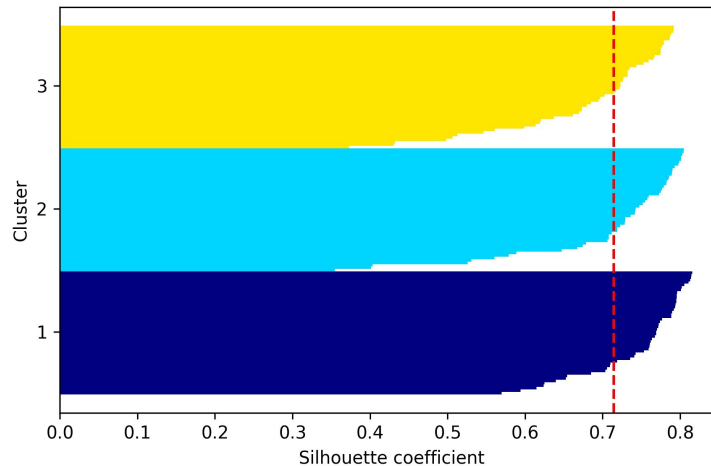
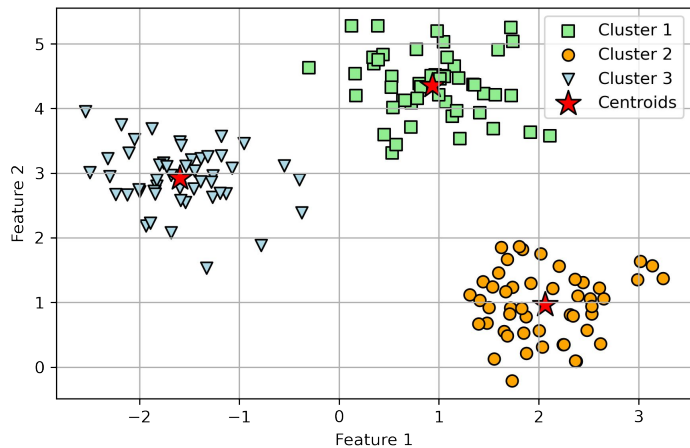
How to assess the quality of clustering?

Silhouette score

1. Calculate the **cluster cohesion**, $a^{(i)}$, as the average distance between an example, $x^{(i)}$, and all other points in the same cluster.
2. Calculate the **cluster separation**, $b^{(i)}$, from the next closest cluster as the average distance between the example, $x^{(i)}$, and all examples in the nearest cluster.
3. Calculate the silhouette, $s^{(i)}$, as the difference between cluster cohesion and separation divided by the greater of the two, as shown here:

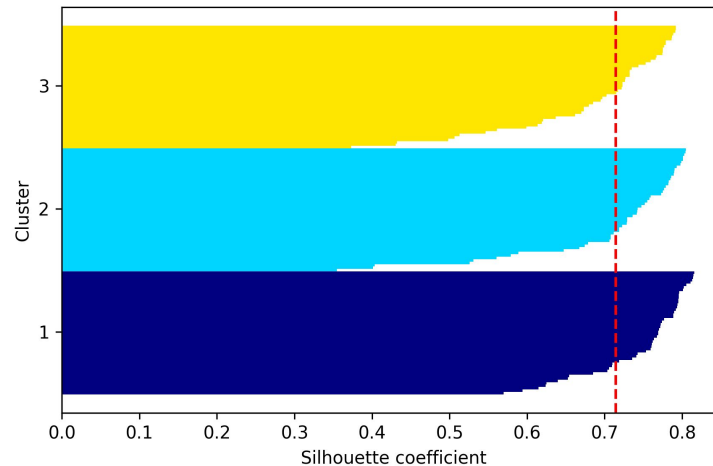
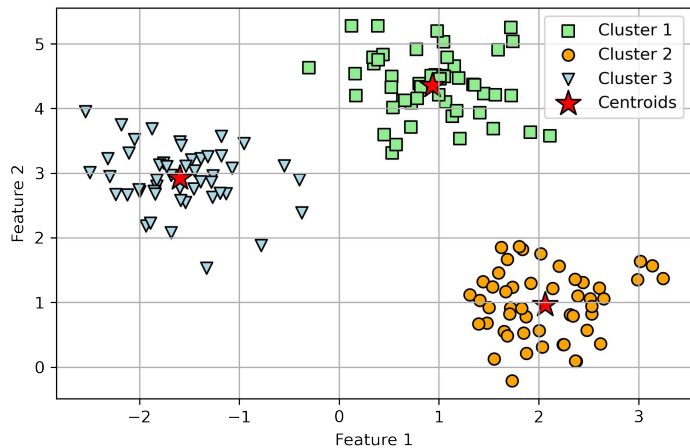
$$s^{(i)} = \frac{b^{(i)} - a^{(i)}}{\max\{b^{(i)}, a^{(i)}\}}$$

How to assess the quality of clustering?

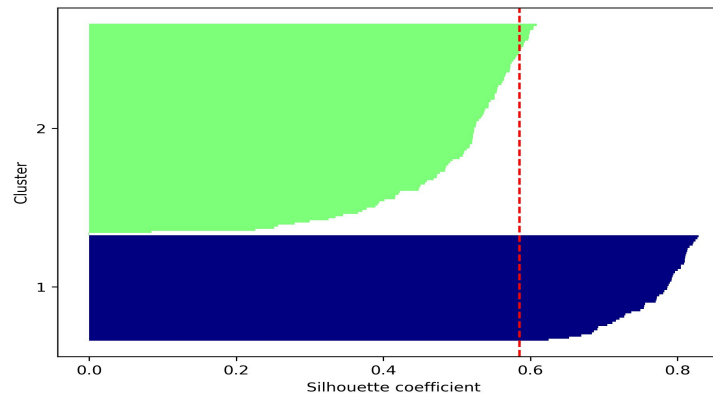
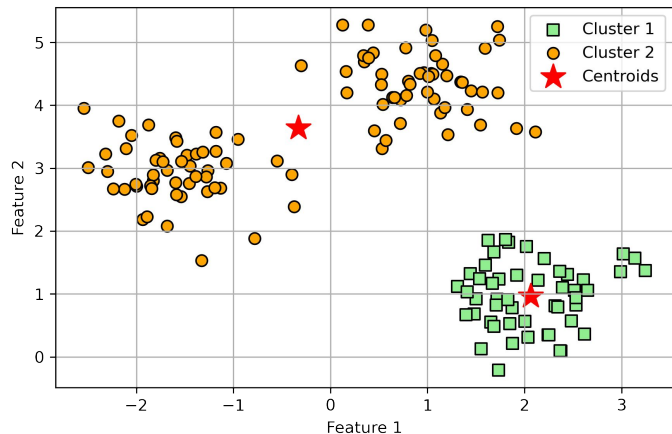


Good clustering:
Silhouette coefficients are not close to 0 and are approximately equally far away from the average silhouette score.

How to assess the quality of clustering?



Good clustering:
Silhouette coefficients are not close to 0 and are approximately equally far away from the average silhouette score.

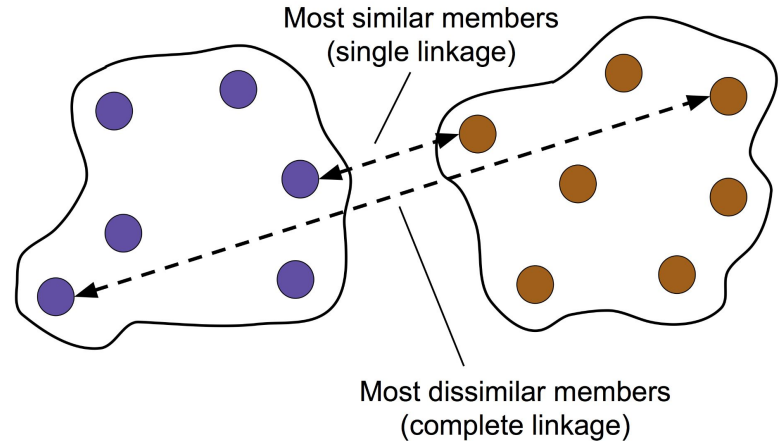


Bad clustering:
Silhouettes have visibly different lengths and widths.

Hierarchical Agglomerative Clustering

Agglomerative clustering starts with each example as an individual cluster and merge the closest pairs of clusters until only one cluster remains.

1. Compute a pairwise distance matrix of all examples.
2. Represent each data point as a single cluster.
3. Merge the two closest clusters based on the distance between the most dissimilar (complete linkage) or most similar (single linkage) members.
4. Update the cluster linkage matrix.
5. Repeat steps 2-4 until one single cluster remains.



Hierarchical Agglomerative Clustering

Let's understand this with an example.

Dataset

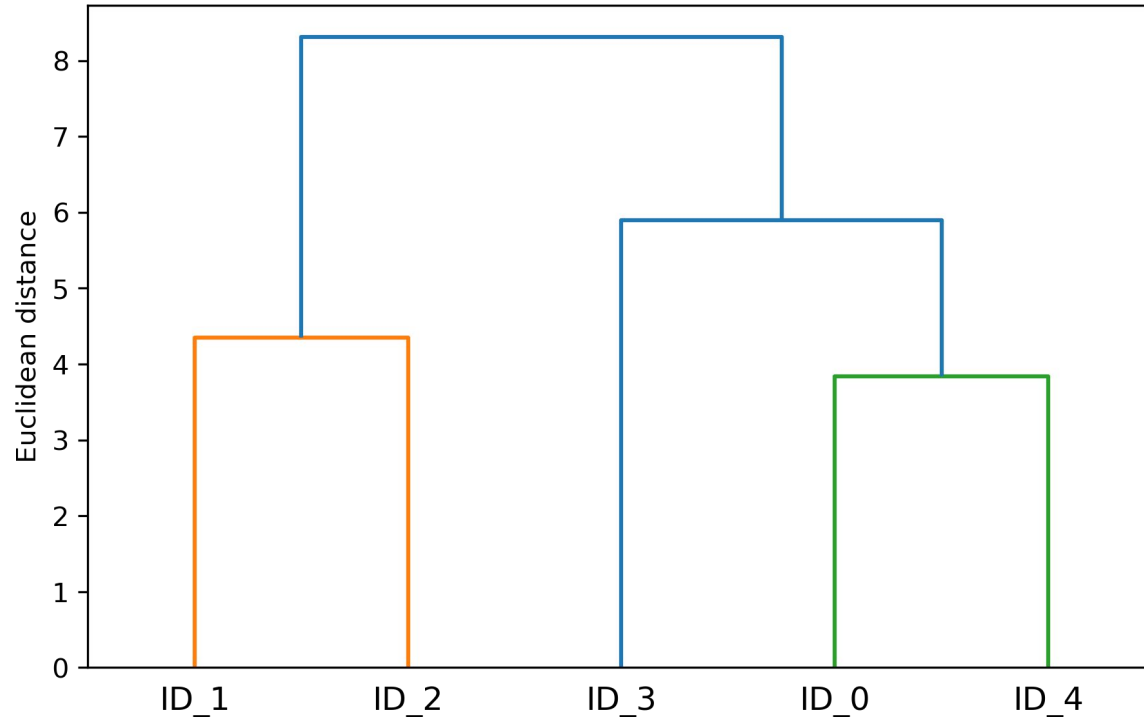
	X	Y	Z
ID_0	6.964692	2.861393	2.268515
ID_1	5.513148	7.194690	4.231065
ID_2	9.807642	6.848297	4.809319
ID_3	3.921175	3.431780	7.290497
ID_4	4.385722	0.596779	3.980443

Distance Matrix

	ID_0	ID_1	ID_2	ID_3	ID_4
ID_0	0.000000	4.973534	5.516653	5.899885	3.835396
ID_1	4.973534	0.000000	4.347073	5.104311	6.698233
ID_2	5.516653	4.347073	0.000000	7.244262	8.316594
ID_3	5.899885	5.104311	7.244262	0.000000	4.382864
ID_4	3.835396	6.698233	8.316594	4.382864	0.000000

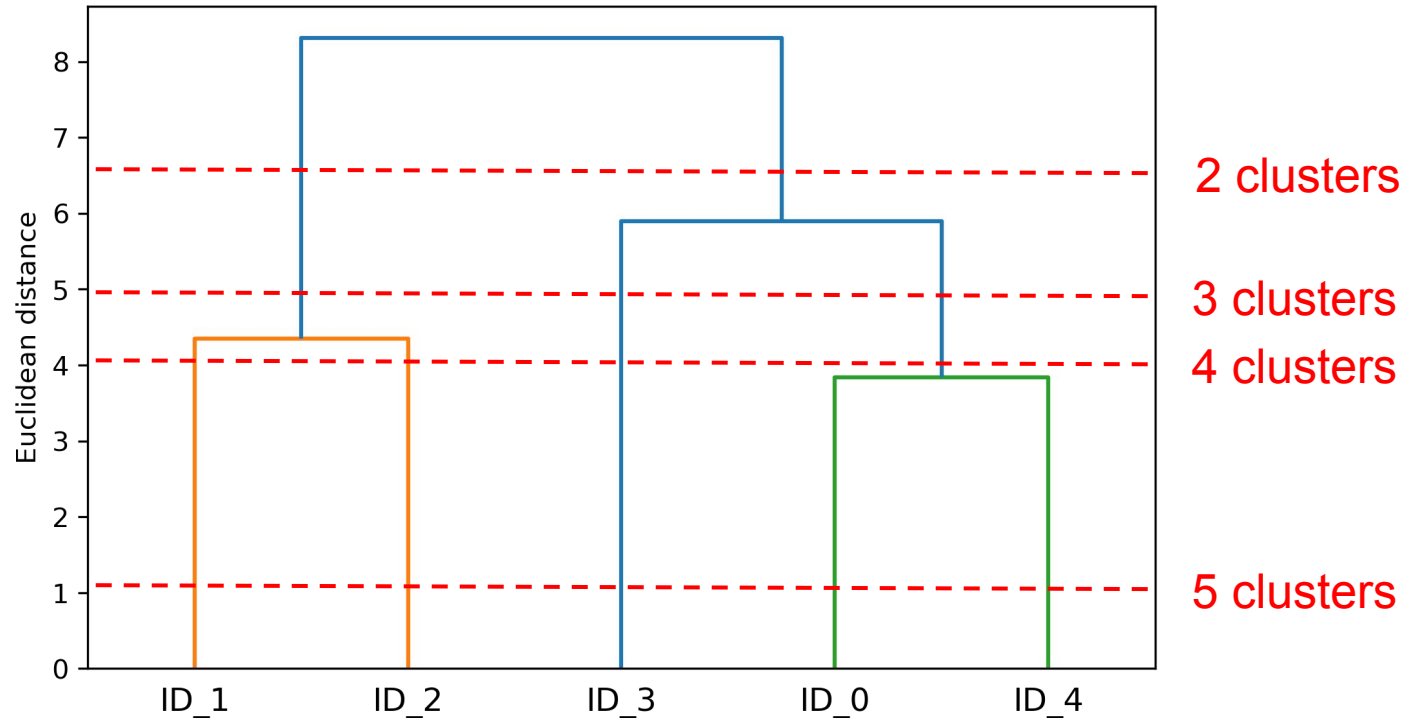
Hierarchical Agglomerative Clustering

Now create clusters by merging the closest clusters/datapoints



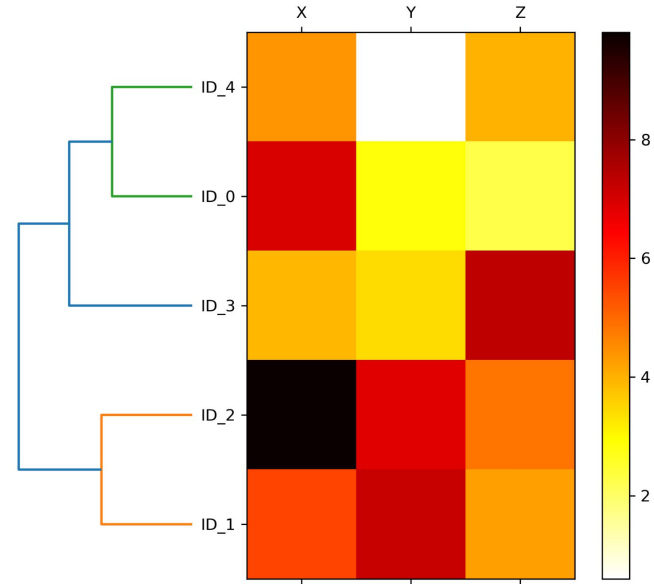
Hierarchical Agglomerative Clustering

As we know the full hierarchy (dendrogram) we can choose the appropriate number of clusters by setting the cut-off distance



Hierarchical Agglomerative Clustering

Hierarchical clustering dendrograms are often used in combination with a heat map, which allows us to represent the individual values in the data array or matrix containing our training examples with a color code.

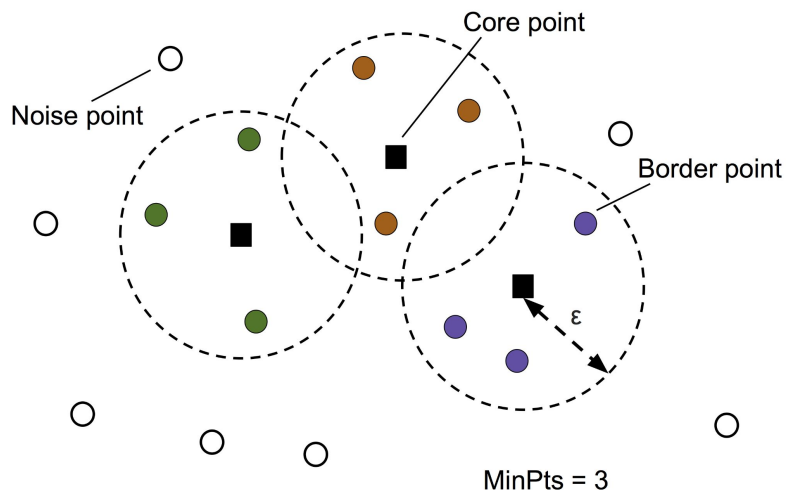


Density Based Clustering (DBSCAN)

Density-based Spatial Clustering of Applications with Noise (DBSCAN)

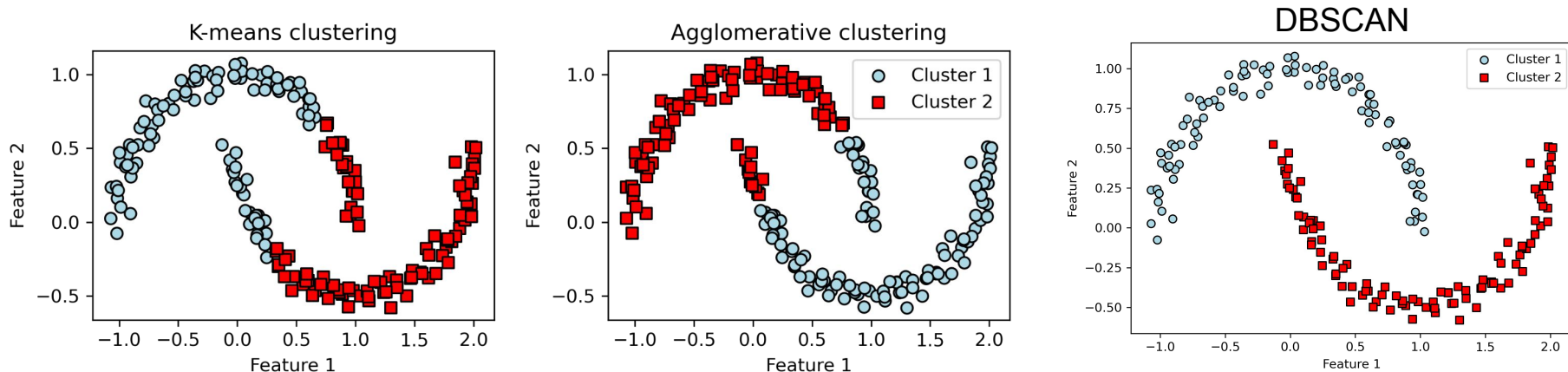
does not make assumptions about spherical clusters like k-means, nor does it partition the dataset into hierarchies that require a manual cut-off point.

1. A point is a core point if at least a specified number (**MinPts**) of neighboring points fall within the specified **radius ϵ**
2. A border point is a point that has fewer neighbors than MinPts within ϵ , but lies within ϵ
3. All other points that are neither core nor border points are considered noise points.
4. Form a separate cluster for each core point or connected group of core points. (Core points are connected if they are no farther away than ϵ .)
5. Assign each border point to the cluster of its corresponding core point.



Density Based Clustering (DBSCAN)

DBSCAN can perform superior clustering when the distribution of the data points are not spherical.

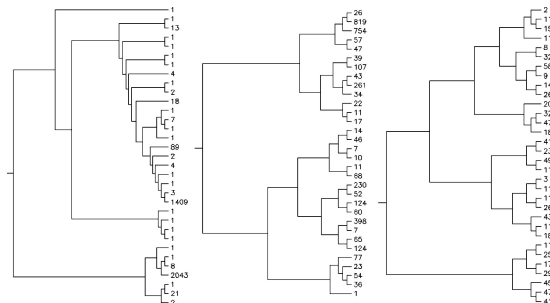


Works well with non linearly separable data points in the feature space.

Automatically determines the optimal number of clusters.

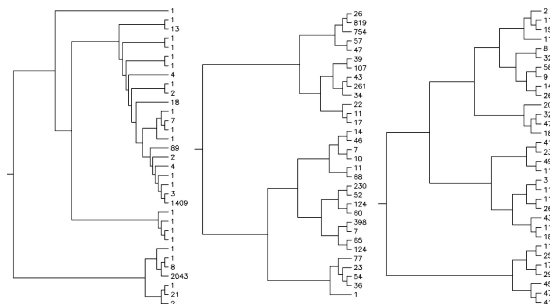
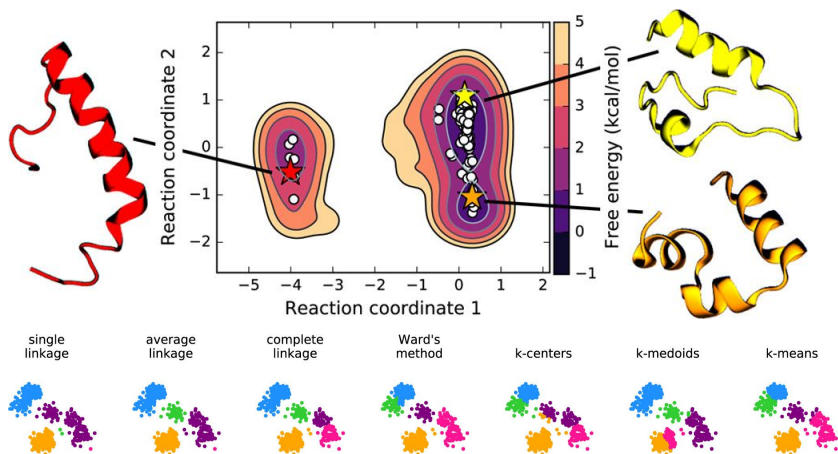
Throws out data that it cannot cluster

Clustering examples on molecular data



Clustering Molecular Dynamics Trajectories: 1. Characterizing the Performance of Different Clustering Algorithms. Shao et al. *Journal of Chemical Theory and Computation* **2007** 3 (6), 2312-2334
DOI: 10.1021/ct700119m

Clustering examples on molecular data



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Ward Clustering Improves Cross-Validated Markov State Models of Protein Folding

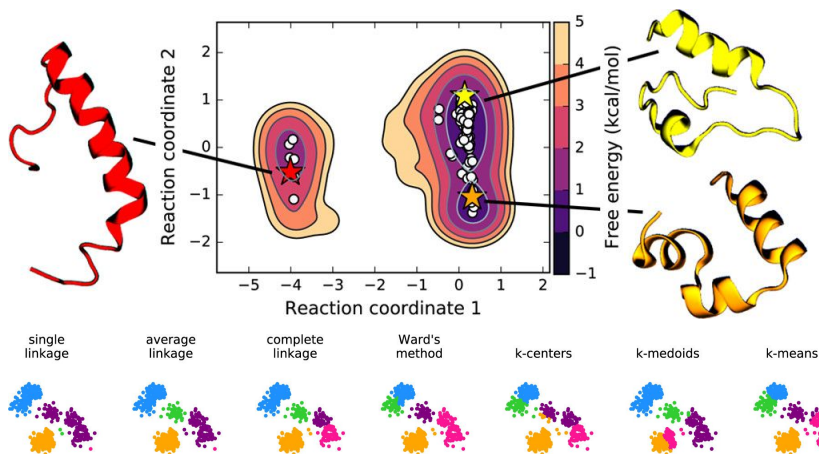
Brooke E. Husic and Vijay S. Pande

Journal of Chemical Theory and Computation

2017 13 (3), 963-967

DOI: 10.1021/acs.jctc.6b01238

Clustering examples on molecular data



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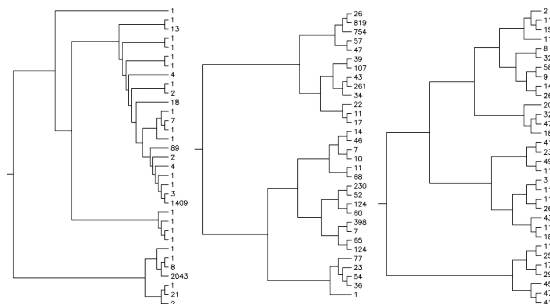
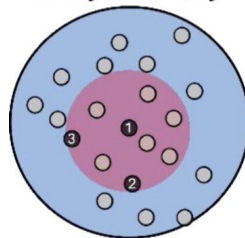
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n -ary similarity



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k-Means NANI: An Improved Clustering Algorithm for Molecular Dynamics Simulations

Lexin Chen, Daniel R. Roe, Matthew Kochert, Carlos Simmerling, and Ramón Alain Miranda-Quintana

Journal of Chemical Theory and Computation **2024** 20 (13), 5583-5597

DOI: 10.1021/acs.jctc.4c00308

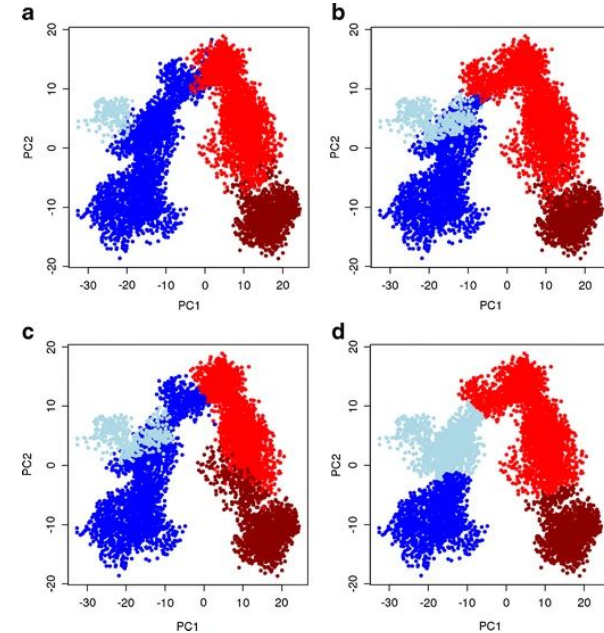
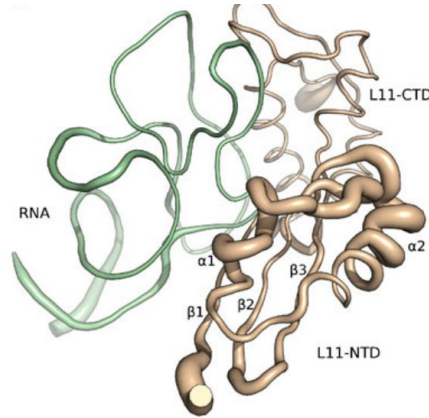
Clustering after dimensionality reduction

Distance measure:
Euclidean distance

$$d(\mathbf{x}, \mathbf{y})^2 = \sum_{j=1}^m (x_j - y_j)^2 = \|\mathbf{x} - \mathbf{y}\|_2^2$$

Euclidean distance is not meaningful in high dimension (e.g. MD simulation data).

Perform dimensionality reduction (e.g. PCA) before clustering.



Wolf, A., Kirschner, K.N. **Principal component and clustering analysis on molecular dynamics data of the ribosomal L11-23S subdomain.** J Mol Model 19, 539–549 (2013).
<https://doi.org/10.1007/s00894-012-1563-4>